

HEMAVAN TARNABY, Sweden

WMO: 021010

Lat: 65.808N

Lon: 15.085E

Elev: 1503

StdP: 13.92

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.6	-20.3	-31.0	1.0	-25.6	-25.3	1.4	-20.3	33.9	19.3	29.6	22.0	1.1	90	0.742

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.1	73.2	59.5	69.2	57.5	65.6	55.9	61.6	69.6	59.1	66.5	56.9	63.8	7.0	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
57.9	75.6	65.7	55.6	69.6	62.6	53.5	64.3	60.3	28.1	69.6	26.4	66.5	25.0	64.1	70.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.1	19.5	16.3	-31.2	77.6	4.9	4.1	-34.8	80.5	-37.7	82.8	-40.4	85.1	-44.0	88.1		
(5)			-31.6	63.4	4.8	3.5	-35.1	66.0	-37.9	68.1	-40.6	70.1	-44.1	72.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	32.8	13.9	14.3	20.1	29.8	39.9	47.9	54.2	51.9	44.2	34.4	23.8	17.8		
	DBStd	17.25	14.93	14.61	11.11	7.82	5.91	5.78	5.68	5.14	5.69	7.00	10.99	14.81		
	HDD50	6598	1120	1000	927	606	317	109	23	38	188	485	787	998		
	HDD65	11760	1585	1420	1392	1056	777	512	337	407	625	949	1237	1463		
	CDD50	316	0	0	0	0	6	47	153	97	13	0	0	0		
	CDD65	2	0	0	0	0	0	0	2	0	0	0	0	0		
	CDH74	66	0	0	0	0	1	4	56	5	0	0	0	0		
	CDH80	5	0	0	0	0	0	0	5	0	0	0	0	0		
	WSAvg	5.8	6.3	6.2	6.4	5.8	6.3	6.3	5.5	5.0	5.6	5.6	5.4	5.8		
	PrecAvg	28.4	2.7	2.0	2.0	1.2	1.3	1.9	3.3	2.9	2.8	2.8	2.6	2.8		
(15) Precipitation	PrecMax	41.5	9.3	7.7	6.3	2.6	3.8	3.8	6.8	5.1	6.2	7.6	6.0	9.4		
	PrecMin	16.4	0.5	0.4	0.2	0.4	0.2	0.2	0.5	0.5	1.2	0.4	0.8	0.7		
	PrecStd	6.1	1.9	1.5	1.6	0.5	0.7	1.0	1.6	1.2	1.1	1.5	1.5	1.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.5	39.1	41.3	52.0	68.8	73.8	79.2	74.3	63.9	52.5	44.6	40.2		
		MCWB	36.9	36.9	35.8	42.5	52.9	57.5	64.4	60.2	53.9	47.7	42.7	38.1		
	2%	DB	37.2	37.1	38.5	47.0	62.4	68.9	75.1	70.1	58.6	49.2	41.1	38.2		
		MCWB	35.1	35.3	35.2	39.3	50.5	54.4	61.0	58.7	52.0	45.8	39.0	36.0		
	5%	DB	34.7	35.0	36.5	43.0	57.0	64.4	70.9	66.5	55.7	46.5	37.9	36.2		
		MCWB	32.8	33.1	33.7	37.7	47.6	52.7	59.4	57.5	50.9	43.6	35.9	34.0		
	10%	DB	32.0	32.6	34.4	40.3	51.8	60.3	66.9	62.7	53.5	44.3	36.0	34.1		
		MCWB	30.2	30.8	32.1	36.1	44.0	50.9	57.5	55.5	49.3	41.7	34.2	32.2		
	0.4%	WB	37.6	37.3	37.3	43.1	54.8	59.1	66.2	63.2	55.7	48.9	43.0	38.7		
		MCDB	39.0	38.7	39.9	51.2	67.1	68.6	75.8	70.7	60.8	51.8	44.3	39.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	35.0	35.4	35.5	40.3	51.1	56.0	63.1	60.4	53.3	45.9	39.0	36.2		
		MCDB	36.6	37.0	37.7	45.7	60.7	65.1	71.6	67.5	57.3	48.5	41.0	37.8		
	5%	WB	33.0	33.3	33.9	38.3	47.7	53.9	60.5	58.3	51.4	43.8	36.3	34.3		
		MCDB	34.6	34.8	35.9	42.2	56.4	62.3	68.4	64.9	55.0	46.2	37.7	35.8		
	10%	WB	30.2	30.9	32.1	36.5	44.7	51.9	58.2	56.1	49.7	41.8	34.3	32.4		
		MCDB	32.0	32.5	34.2	39.9	51.0	59.7	65.6	61.8	53.1	44.1	35.7	33.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	14.4	15.2	17.3	15.1	14.6	15.0	17.1	16.4	13.4	9.2	9.7	12.6		
		MCDBR	10.9	12.0	13.8	19.1	26.9	25.1	26.6	24.7	17.7	10.7	7.7	9.7		
		MCWBR	10.1	10.9	11.2	12.6	16.0	12.6	14.0	14.4	11.5	8.4	6.8	8.8		
	5% WB	MCDBR	11.4	11.8	12.2	15.5	25.9	21.9	23.3	21.9	15.4	10.5	7.6	10.0		
		MCWBR	10.8	10.8	10.3	10.4	15.8	11.7	13.4	13.7	10.6	8.7	6.8	9.5		
(40) Clear-Sky Solar Irradiance	taub		0.174	0.244	0.278	0.304	0.315	0.317	0.329	0.315	0.291	0.265	N/A	0.177		
	taud		1.928	2.173	2.222	2.246	2.355	2.498	2.502	2.525	2.571	2.383	N/A	1.937		
	Ebn at Noon		95	204	252	273	282	285	277	271	254	199	N/A	17		
	Edh at Noon		6	16	26	32	32	29	28	25	19	13	N/A	2		
(44) All-Sky Solar Radiation	RadAvg		27	161	594	1184	1519	1538	1437	1079	581	211	51	6		
	RadStd		2	34	75	96	104	142	176	116	71	28	6	0		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (34 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page